# CS 361 – Project 1

***You may work in pairs for this assignment (recommended!).*** *If you do work in pairs, sign up for a group on Canvas, and only one partner needs to submit.*

Your goal in this assignment is to use an existing app or website to reverse engineer a database design. Choose an app or website that you like to use or that you use relatively often (the more familiar you are the easier it will be to reason about what is needed), and reverse-engineer the underlying data model. The model should encompass all data needed to carry out the operations and functions in the application, so be careful to consider all data you are able to see and interact with as a user. Also pay attention to how a user's data may change over time and what is likely being tracked over time by the application's owners.

**Requirements** You must create and submit the following:

1. **Data Requirements**: You must write the data requirements for the chosen application. This includes describing the data that the database must keep track of as well as functional requirements of what the users of the application should be able to do with the data. Essentially, you are producing a document that can be used to conduct the conceptual phase of design: your document should be detailed enough that one could use it to create an entity-relationship model without ***any*** familiarity with the chosen application.

2. **Entity-Relationship Diagram**: You must produce a digital entity-relationship diagram representing the data for the application. Your diagram must meet the requirements articulated in part 1: the data needed must be present, and the functional requirements must be supported. You may use software of your choice to create your digital diagram. Some recommendations include Draw.io, DiagramDesigner, Lucidchart Diagrams, Visio (Microsoft), Canva, etc.). You **MUST** follow the E/R diagram conventions we used in lecture. Your diagram may be accompanied by written descriptions explaining or clarifying aspects of your model.

3. **Relational Schema**: You must create a relational schema derived from your entity-relationship diagram. This relational schema should show the relation names, attribute names and the primary key for each relation. Choose distinctive relation and attribute names (i.e., differentiate similar attributes from different entities).

4. **Database DDL:** You must write a data definition script in a file called **`p1ddl.sql`** implementing the creation of your relational schema in a MySQL/MariaDB database. Your script must also include drop commands so the script can be re-run to re-create the database multiple times, if necessary.

5. **Database Data:** You must create a SQL script called **`p1data.sql`** with commands to populate your database with sample data. You should insert at least four rows of data into each table (unless table ***should*** have less than four rows); insert more than four rows if necessary and/or desirable to demonstrate your final database. Choose varied, representative data of the requirements. Demonstrate that ultimately your design can store the data that it is supposed to.

Submit all the above in a single zipped folder to Canvas prior to the deadline.